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- Div: B
- Practical Batch: B2
- Roll No: 46
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- Practical No: 01
- Practical Name: Solve Any problem using depth first search

```
def dfs(start, adj):
    visited = [0] * len(adj)
    result = []
    stack = []

    stack.append(start)

    while stack:
        node = stack.pop()

        if visited[node] == 0:
            visited[node] = 1
            result.append(node)

            # Add neighbors in reverse order
            # to get the same traversal as recursive DFS
            for n in reversed(adj[node]):
                if visited[n] == 0:
                    stack.append(n)

    return result

num_of_nodes = 7

adj = [
    [],
    [2, 3],
    [1, 4, 5],
    [1, 6, 7],
    [2],
    [2],
    [3],
    [3]
]

result = dfs(1, adj)

print("DFS Traversal:", result)
```

DFS Traversal: [1, 2, 4, 5, 3, 6, 7]